EE 121B: Principles of Semiconductor Device Design

Lecture 10:

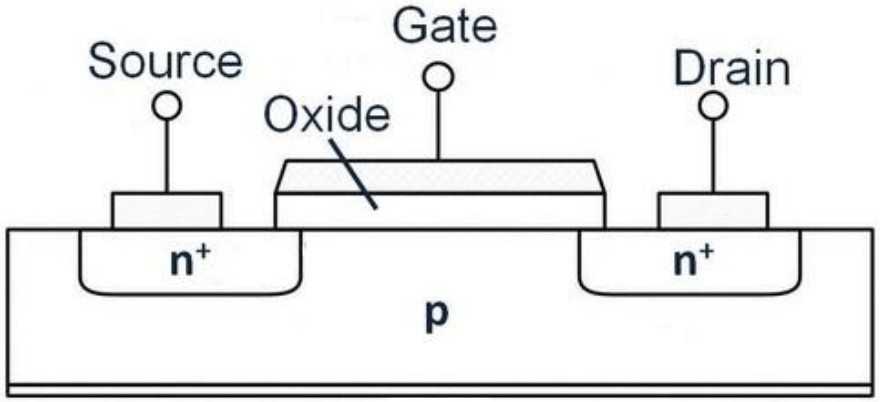
pn Junction Diodes

Sam Emaminejad

Course Overview

1. Semiconductor Properties
2. Metal-Semiconductor Contacts
3. P-N Junction
4. MOS Devices
5. Bipolar Junction Transistor

Metal-Oxide-Semiconductor (MOS)
Field-Effect Transistor (FET)



http://qph.is.quoracdn.net/main-qimg-38a99b4f215e161bf31964795cfce7d2?convert_to_webp=true

Lecture 10

OUTLINE

- pn Junction Diodes (cont'd)
  - Summary of important concepts
  - Diode applications
    - Temperature sensor
    - Varactor diodes
    - Optoelectronic diodes

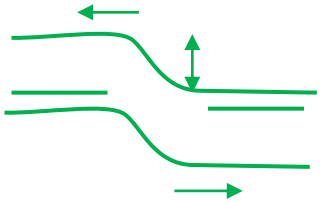
Reading: Pierret 9; Hu 4.12-4.15

Summary of Important Concepts

- Under **forward** bias, minority carriers are **injected** into the quasi-neutral regions of the diode.
- The current flowing across the junction is comprised of hole and electron components.
  - If the junction is asymmetrically doped (*i.e.* it is "one-sided") then one of these components will be dominant.

Diode current: $I = qA\left(\frac{D_n n_i^2}{L_n N_A} + \frac{D_p n_i^2}{L_p N_D}\right)(e^{\frac{qV_A}{kT}} - 1)$

Electron diffusion in p-side Hole diffusion in n-side



- In a long-base diode, the injected minority carriers recombine with majority carriers within the quasi-neutral regions.

Visible light (photons) is emitted if the semiconductor has a direct bandgap (III-IV semiconductor)

$$I = qA\left(\frac{D_n n_i^2}{L_n N_A} + \frac{D_p n_i^2}{L_p N_D}\right)(e^{\frac{qV_A}{kT}} - 1)$$

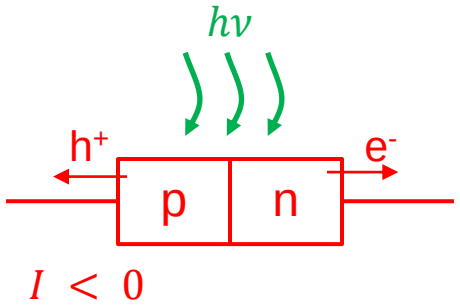
Electron diffusion in p-side Hole diffusion in n-side

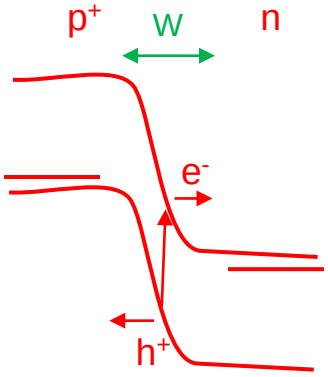
- The ideal diode equation stipulates the relationship between $J_N(-x_p)$ and $J_P(x_n)$:

$$\frac{J_N(-x_p)}{J_P(x_n)} = \frac{D_n L_p N_D}{D_p L_n N_A}\left(\frac{{n_i}^2{}_{p-side}}{{n_i}^2{}_{n-side}}\right)$$

➢ For example, if holes are forced to flow across a forward-biased junction, then electrons must also be injected across the junction.

- Under **reverse** bias, minority carriers are **collected** into the quasi-neutral regions of the diode.
  - Minority carriers generated within a diffusion length of the depletion region diffuse into the depletion region and then are swept across the junction by the electric field.





L_n, L_p > 1μm

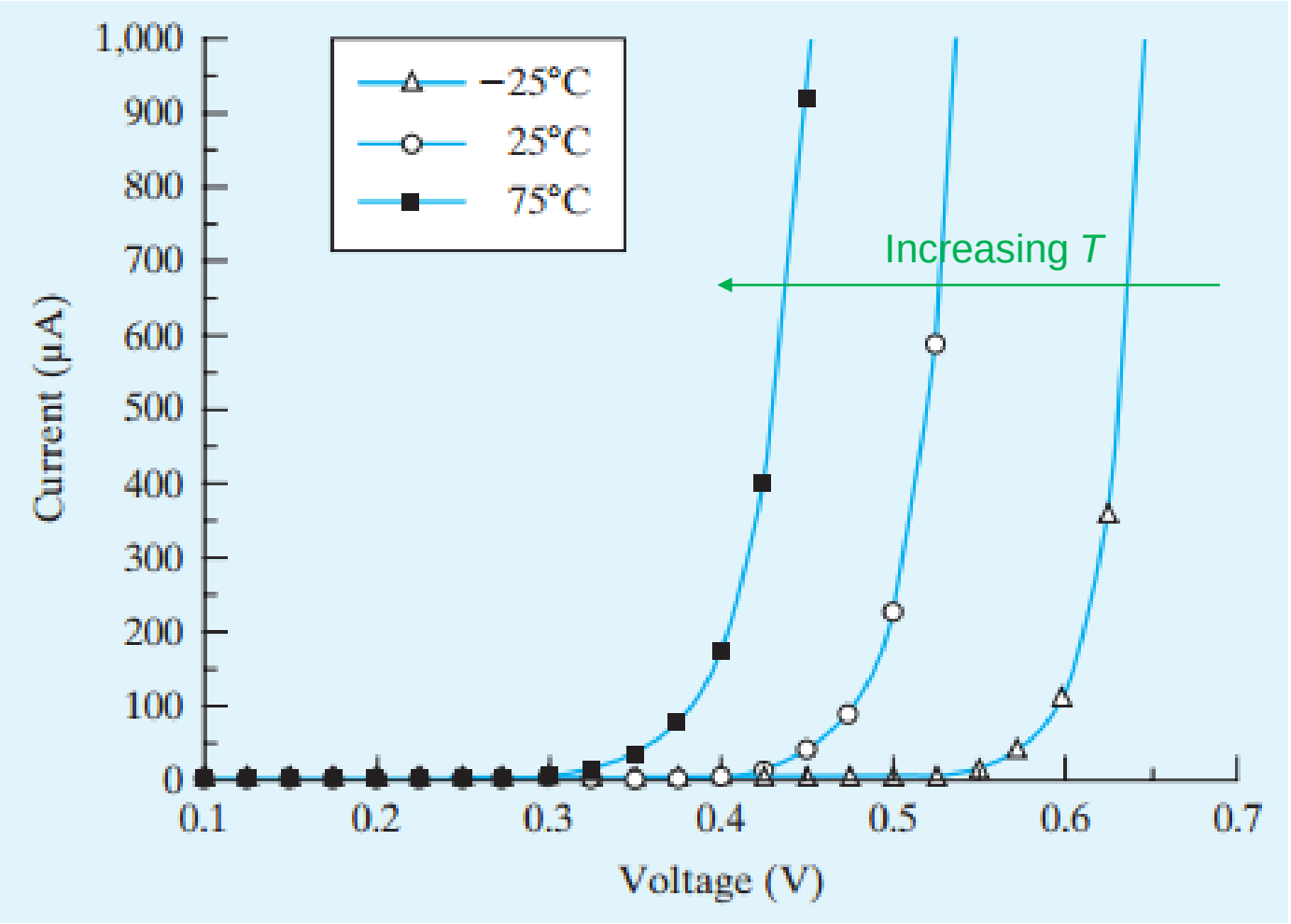
$$qV_{bi} < E_G$$

$$q(V_{bi} - V_A) > E_G > qV_{bi}$$

→The negative current flowing in a reverse-biased diode depends on the rate at which minority carriers are supplied from the quasi-neutral regions.

- Electron-hole pair generation within the depletion region also contributes negative diode current.

pn Junction as a Temperature Sensor



C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 4-21

Varactor Diode

Reverse-biased diode
$-VA = V_r$

- Voltage-controlled capacitance
  - Used in oscillators and detectors
    (*e.g.* FM demodulation circuits in your radios)
  - Response changes by tailoring doping profile:

Tune C_j and hence the oscillation frequency

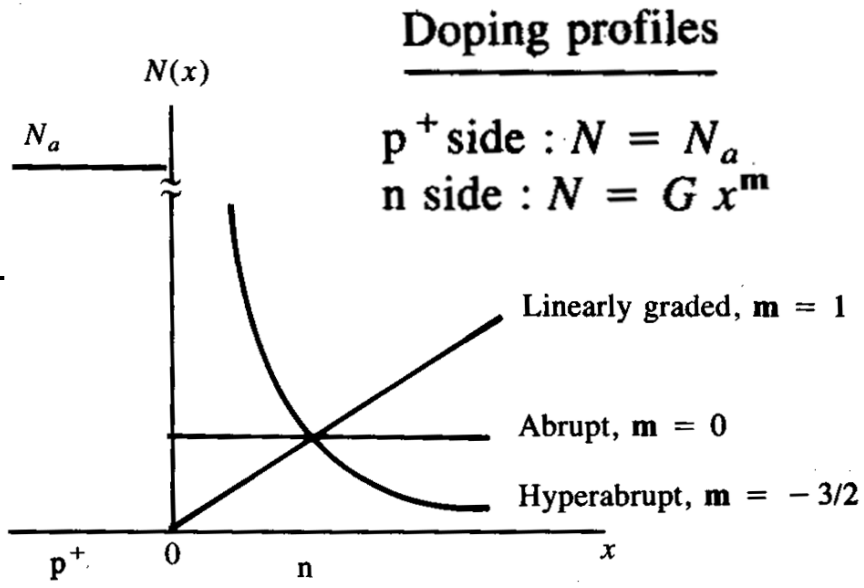
$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$C_j \propto V_r^{-n}$$

for

$$V_r >> V_{\text{bi}}$$

$$n = \frac{1}{m+2}$$



Preview: Optoelectronics

- The relationship between the optical wavelength (λ) and the associated photon energy ($E_{ph} = hv$) is given by:

$$\lambda = \frac{c}{v} = \frac{hc}{hv} = \frac{hc}{E_{ph}}$$

h: Planck's constant
v: wave frequency
c: speed of light

- If λ is expressed in 'µm' and in E_{ph} 'eV', then:

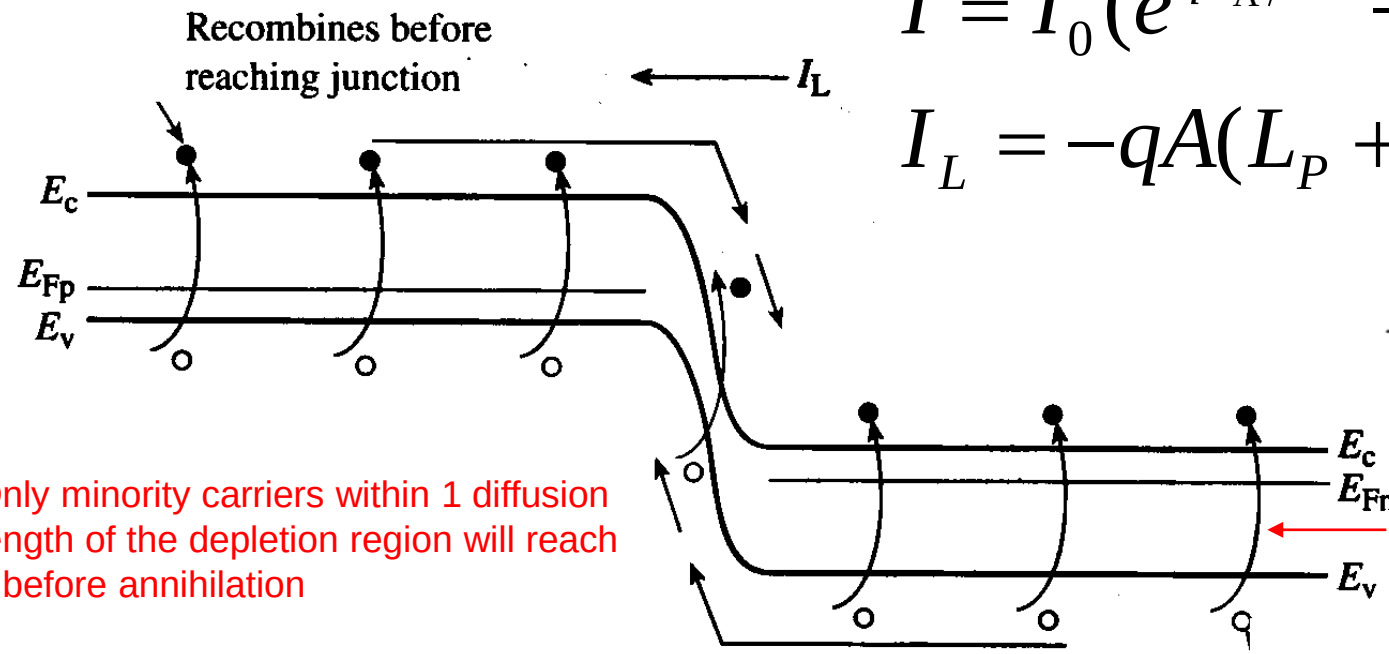
$$\lambda = \frac{1.24}{E_{ph}} \qquad E_{ph} = \frac{1.24}{\lambda}$$

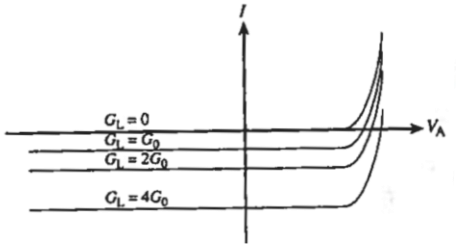
This is helpful to contextualize photon energy in relation to the semiconductor band gap energy.

Optoelectronic Diodes

$$I = I_0(e^{qV_A/kT} - 1) + I_L$$

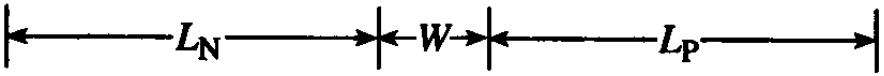
$$I_L = -qA(L_P + W + L_N)G_L$$

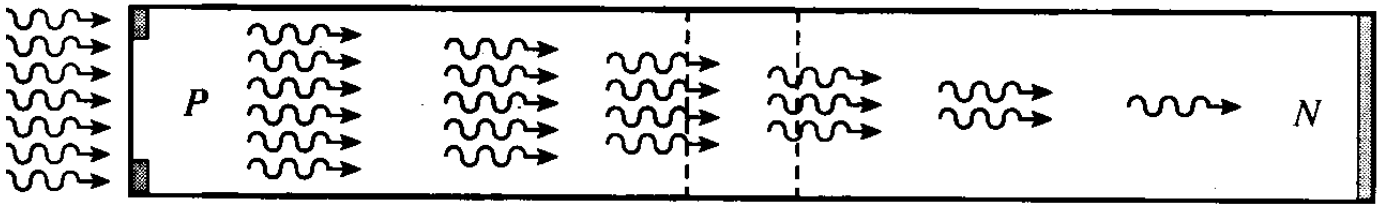




Only minority carriers within 1 diffusion length of the depletion region will reach it before annihilation

Electron-hole pair generation due to light





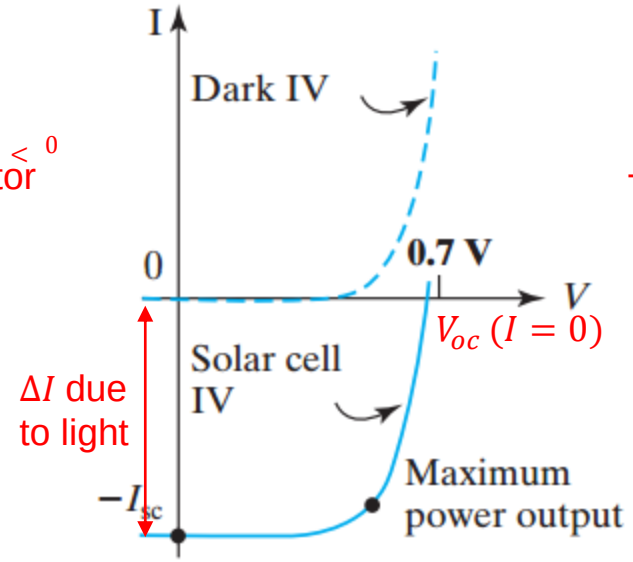
R.F. Pierret, *Semiconductor Fundamentals*, Figure 9.2

Open Circuit Voltage, V_{OC}

$$V_{oc} = V_A\big|_{I=0} = \frac{kT}{q}\ln\left[\frac{L_p + W + L_n}{\left(L_p/\tau_p\right)p_n + \left(L_n/\tau_n\right)n_p}G_L + 1\right]$$

Operate with $V_A > 0$ for solar cell ($IV < 0$)

Operate with $V_A < 0$ for photodetector



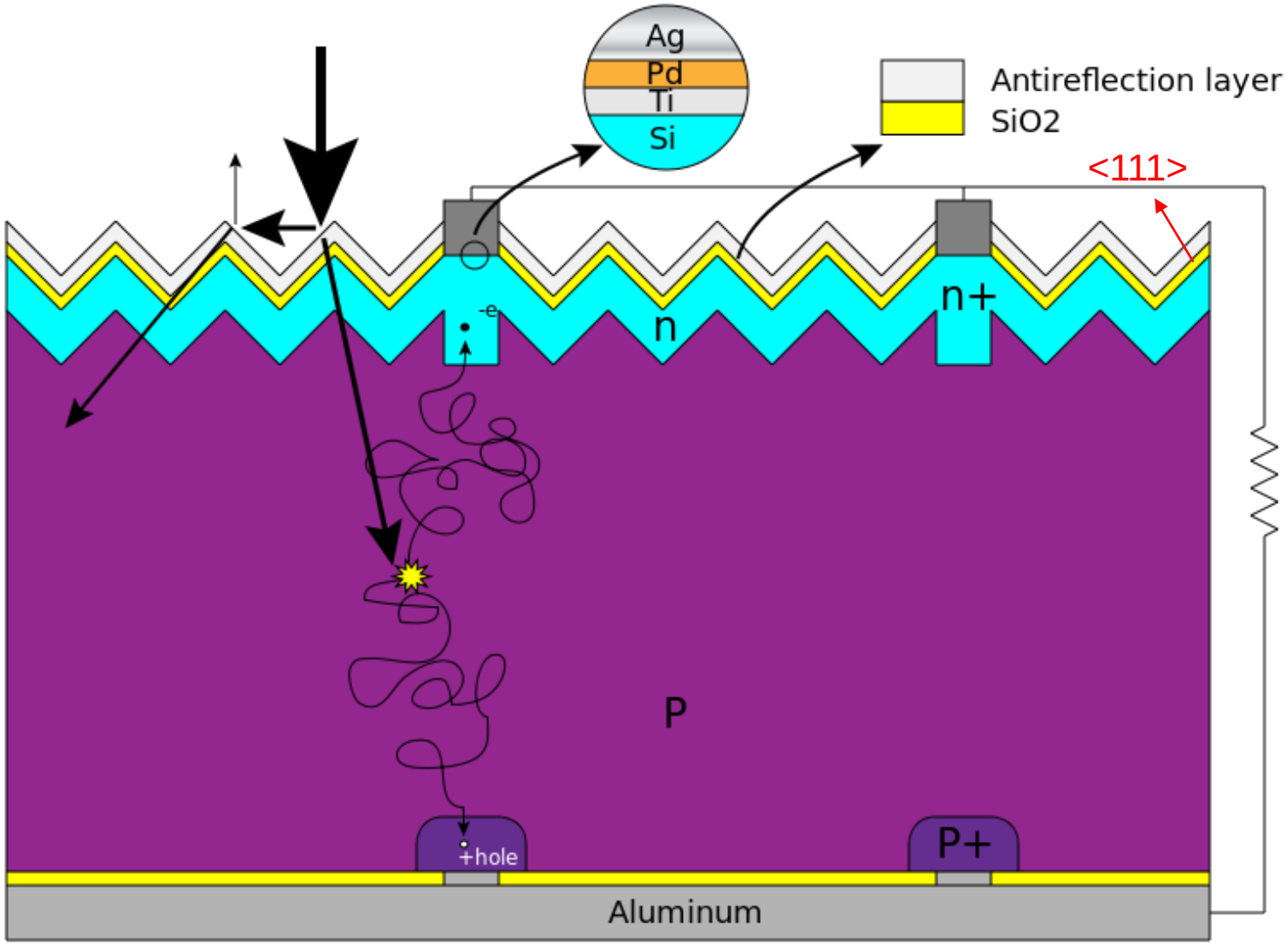
Typically, $V_{oc} \cong 0.6\ \text{V}, I_{sc} \cong 50\ \text{mA/cm}^2$ for Si solar cell

$$Fill\ factor \equiv \frac{P_{out}}{I_{sc}V_{oc}} \sim 75\%$$

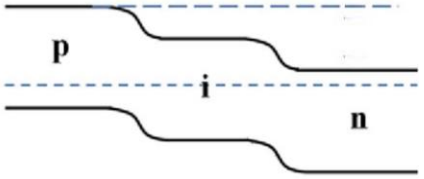
C. C. Hu, *Modern Semiconductor Devices for ICs*, Figure 4-25(b)

Solar Cell Structure

Cyferz at en.wikipedia



p-i-n Photodiodes

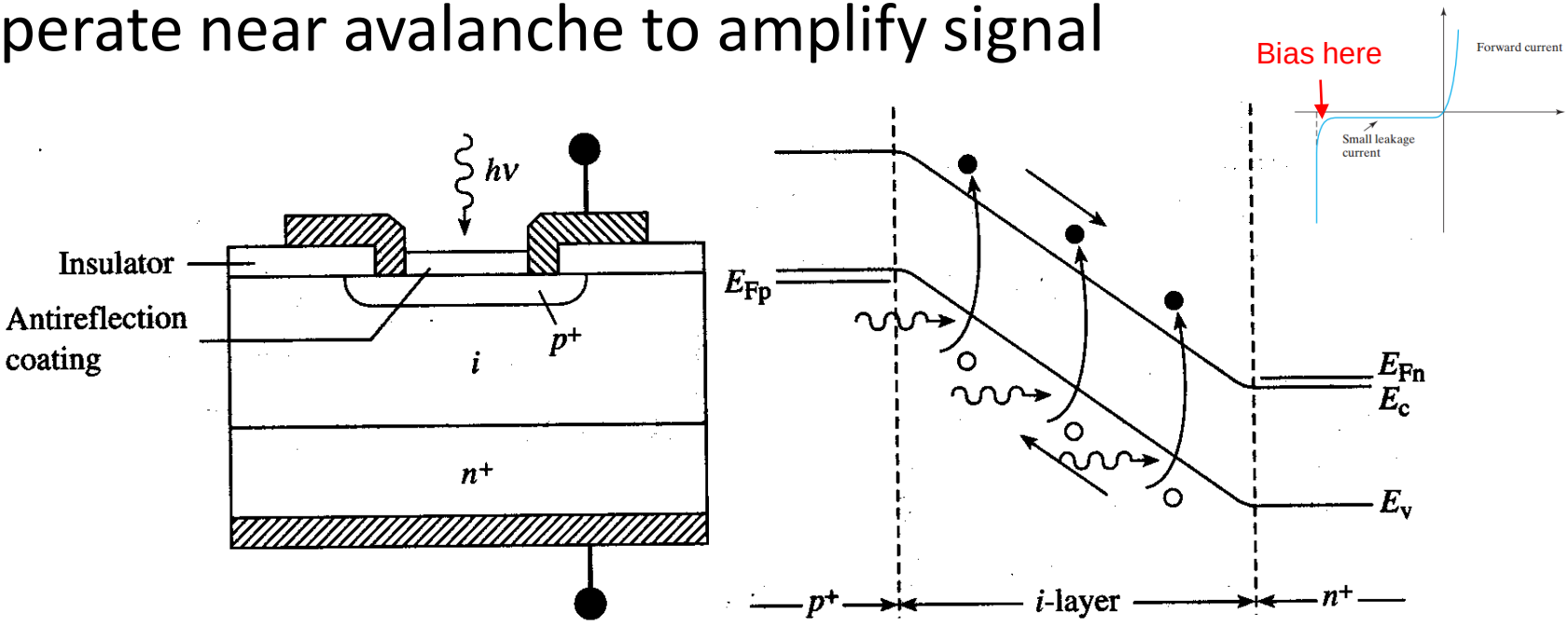


- $W \cong W_{\text{i-region}}$, so most carriers are generated in the depletion region *vs*. in the quasi-neutral regions as in a pn diode

  → faster response time (~10 GHz operation)

  Since generated carriers do not need to diffuse to the depletion region before they are collected

- Operate near avalanche to amplify signal



R.F. Pierret, *Semiconductor Fundamentals*, Figure 9.5

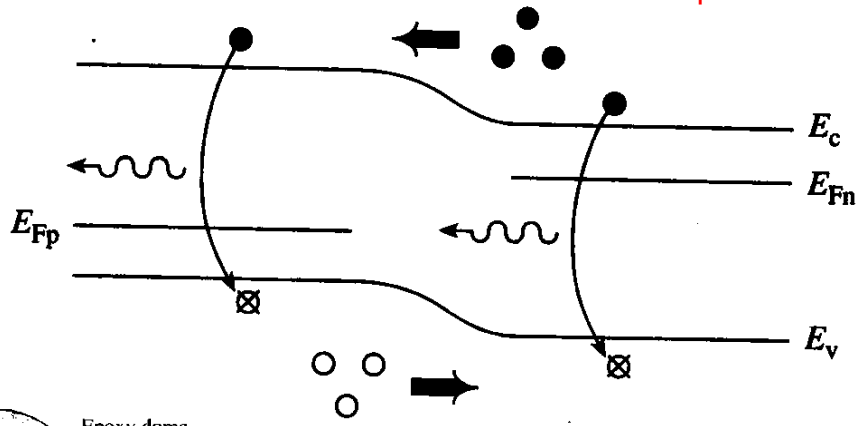
- Efficiency
- Compact size
- Long lifetime (~100,000 hrs)

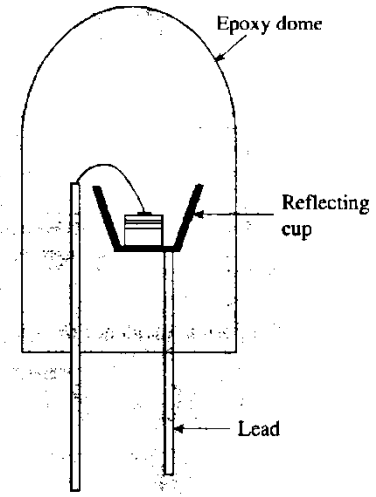
Light Emitting Diodes (LEDs)

- LEDs are made with compound semiconductors (direct bandgap)

R.F. Pierret, *Semiconductor Fundamentals*, Figure 9.13

Photon is emitted when electron-hole pair recombines





R.F. Pierret, *Semiconductor Fundamentals*, Figure 9.15

$$\lambda = \frac{1.24}{E_G\ (\text{eV})}\ \text{in}\ \mu\text{m}$$

Visible spectrum:

Red: $\lambda = 650$ nm
Green: $\lambda = 550$ nm
Blue: $\lambda = 450$ nm

Semiconductor	*Color*	*Peak λ(μm)*
$GaAs_{0.6}P_{0.4}$	Red	0.650
$GaAs_{0.35}P_{0.65}$:N	Orange-Red	0.630
$GaAs_{0.14}P_{0.86}$:N	Yellow	0.585
GaP:N	Green	0.565
GaP:Zn-O	Red	0.700
AlGaAs	Red	0.650
AlInGaP	Orange	0.620
AlInGaP	Yellow	0.585
AlInGaP	Green	0.570
SiC	Blue	0.470
GaN	Blue	0.450

White LDE: blue LED + yellow phosphor

Acknowledgement

- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
- Professor Tsu-Jae King Liu EE130/230A notes